USED PAPERS AUG/SEP-2021

Code No: 156AM **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August/September - 2021
CONCURRENT PROGRAMMING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Write Lock one algorithm for two thread problem and give its disadvantages.
b) Examine lock two algorithm for two threads problem. [7+8]

2.a) Explain Peterson Algorithm for achieving mutual exclusion.
b) Prove that Bakery algorithm is dead lock free. [7+8]

3.a) Develop lock based queue and explain.
b) Write about linearliazability, compositional linearliazability and non blocking property. [7+8]

4.a) Compare and Contrast concurrent Vs Sequential object.
b) Analyze synchronized block and locks in Java. [7+8]

5.a) Discuss about wait free consensus protocol.
b) Elaborate compare set operation. [7+8]

6.a) Propose lock free universal algorithm.
b) Compare test-and-set (TAS), test-and-test-and-set (TTAS) operations. [7+8]

7.a) Compare coarse grained Vs fine grained synchronization.
b) Explain how do you represent sets with linked list and illustrate with examples. [7+8]

8.a) Elaborate back off stack with example.
b) Construct unbounded lock free queue. [7+8]

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